

GE 303: Modern Powertrains: Electric, Hybrid, and IC/Aero Engine

Course Code	GE 303
Course Name	Modern Powertrains: Electric, Hybrid, and IC/Aero Engine
L-T-P-C	2-0-1-3
Prerequisites	None
Intended for	UG 3rd, 4th year, and PG students
Mutual Exclusion	None
Approval	61st BoA (Jan 2026); 48th Senate (Mar 2026), in the B.Tech General Engineering with Mechatronics & AI curriculum

Preamble

The course provides an overview of the fundamentals of the modern powertrains, which includes, electric vehicles, hybrid electric vehicles, IC Engines and aero engines/turbines. The learning outcomes will be as follows: Fundamentals and architecture of a typical IC Engine; Advancement in IC Engine technology, Turbocharged, GDI, HCCI, and Flex fuel Engines; Operational fundamentals of an Aero Engine; Types of gas turbines and their working principles; Basics for Hybrid electric powertrain design and their transmission system; Conversion techniques from ICE to HEV; Fundamentals of electric powertrain and design principles for electric vehicles.

The course includes a practical part that will include dismantling & assembly of an IC Engine, dismantling & assembly of an electric scooter, performance and emission experiments on a HEV powertrain, design of planetary gear system for HEV transmission, and overview of MIG-29 aero engine.

Course Modules

- Unit 1: IC Engine Fundamentals:** Explanation of the engine specification of a typical 2-wheeler and a 4-wheeler engine. Fundamental difference between SI and CI Engines. Correlation of piston movement and air standard cycles for 2 Stroke & and 4 Stroke Engines. Understanding the impact of engine displacement, bore, stroke, and bore-to-stroke ratio. (6 Hours)
- Unit 2: Advanced IC Engine Concepts:** Combustion chamber design: SI, DISI, IDI, CI, etc. Valve port and manifold design. Understanding swirl, tumble, supercharging, and turbocharging in engines. Engine block design for air cooled and water-cooled engine. Learning various fuel injection systems: PFI, GDI, Carburetor, etc. (6 Hours)
- Unit 3: Hybrid Electric Powertrains:** Overview of hybrid electric powertrains. Fundamentals of typical HEV powertrains: Parallel hybrid system, series hybrid system, and mixed. Level of hybridization: micro, mild, full, and plug-in HEV. Understanding transmission system of typical HEVs: Honda City HEV, Toyota Hycross HEV, and Yamaha FZ HEV. (6 Hours)
- Unit 4: Electric Powertrain:** Understanding the basic elements of an electric powertrain. Type of motors and principles to select motors for an electric powertrain. Comparative study of all the batteries used in electric powertrains: their advantage & disadvantage. Various charging principles. Status of Charging infrastructure in India. (6 Hours)
- Unit 3: Aero Engine Design:** Fundamentals of gas turbines. Type of aero engines: Turbofan, Turboshift, and Turboprops, their advantages and disadvantages. Fuel injection system for aero engines. Structural elements of aero engines: compressor, combustion chamber, and afterburners, their use and principles. (4 Hours)

Laboratory/Practical/Tutorial Modules: Hands-on Activities (14 Hours)

1. Dismantling and Assembly of a 2 Stroke Engine.
2. Dismantling and Assembly of a 4 Stroke Engine.
3. Dismantling and Assembly of an Electric Scooter.
4. Dismantling and Assembly of an Electric E-Rikshaw.
5. Design & Manufacturing of HEV Planetary Gear System.
6. Performance & Emission testing of a Parallel HEV Powertrain
7. Visual overview of MIG-29 Aero Engine.

Text Books

1. Kevin Hoag & Brian Dondlinger, *Vehicular Engine Design*, Springer.
2. A. Kolchin & V. Demidov, *Design of Automotive Engines*, MIR Publishers.

References

1. Robert Bosch GmbH, *Diesel-Engine Management & Gasoline-Engine Management*.
2. Ehsani Mehrdad, *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*.
3. Iqbal Husain, *Electric and Hybrid Vehicles: Design Fundamentals*.
4. David T. Pratt, Jack Mattingly, and William H. Heiser, *Aircraft Engine Design*.